

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, software, and analytics over the internet (the cloud).

There are three main service models in cloud computing: Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. Platform as a Service (PaaS) offers hardware and software tools over the internet. Software as a Service (SaaS) delivers software applications over the internet on a subscription basis.

Cloud deployment models include Public Cloud, Private Cloud, Hybrid Cloud, and Multi-Cloud.

Major cloud providers include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP).

Benefits of cloud computing include scalability, cost efficiency, flexibility, and high availability.

Cloud computing is widely used in industries such as healthcare, finance, education, and e-commerce.

Security in cloud computing involves data encryption, identity management, and compliance with standards.

Serverless computing allows running code without managing servers.

Containers and Kubernetes are used to deploy and manage applications efficiently.